fields are composed of photons, and gravitational fields are believed to be composed of gravitons, the strong and weak fields also have particulate constituents. The particles of the strong force are called *gluons* and those of the weak force are called W and Z particles. The existence of these force particles was confirmed by accelerator experiments carried out in Germany and Switzerland in the late 1970s and early 1980s.

The field framework also applies to matter. Roughly speaking, the probability waves of quantum mechanics may themselves be thought of as space-filling fields that provide the probability that some or other particle of matter is at some or other location. An electron, for instance, can be thought of as a particle—one that can leave a dot on a phosphor screen, as in Figure 4.4—but it can (and must) also be thought of in terms of a waving field, one that can contribute to an interference pattern on a phosphor screen as in Figure 4.3b.[3] In fact, although I won't go into it in greater detail here,[4] an electron's probability wave is closely associated with something called an *electron field*—a field that in many ways is similar to an electromagnetic field but in which the electron plays a role analogous to the photon's, being the electron field's smallest constituent. The same kind of field description holds true for all other species of matter particles as well.

Having discussed both matter fields and force fields, you might think we've covered everything. But there is general agreement that the story told thus far is not quite complete. Many physicists strongly believe that there is yet a third kind of field, one that has never been experimentally detected but that over the last couple of decades has played a pivotal role both in modern cosmological thought and in elementary particle physics. It is called a Higgs field, after the Scottish physicist Peter Higgs.[5] And if the ideas in the next section are right, the entire universe is permeated by an ocean of Higgs field—a cold relic of the big bang—that is responsible for many of the properties of the particles that make up you and me and everything else we've ever encountered.

Fields in a Cooling Universe